

FAULT CODE 131 - Accelerator Pedal or Lever Position Sensor 1 Circuit - Voltage Above Normal or Shorted to High Source

Warnings and Cautions

CAUTION

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3823996 - female Weather Pack™ test lead and Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Check for sensor supply fault codes.	Fault Code 2185 active?
	STEP 1B. Check for an active fault code.	Fault Code 131 active?
STEP 2.	Check the accelerator pedal or lever position sensor circuit.	
	STEP 2A. Inspect the accelerator pedal position sensor and connector pins.	Dirty or damaged pins?
	STEP 2B. Check the accelerator pedal position sensor supply voltage and return circuit.	Between 4.75-VDC and 5.25-VDC?

STEPS		SPECIFICATIONS
	STEP 2C. Check the circuit response.	Fault Code 132 active and Fault Code 131 inactive?
	STEP 2D. Check the fault codes and verify accelerator pedal or lever condition.	Fault Code 132 active?
STEP 3.	Check the ECM and original equipment harness (OEM) harness.	
	STEP 3A. Inspect ECM and OEM harness connector pins.	Dirty or damaged pins?
	STEP 3B. Check for an open circuit in the OEM harness.	Less than 10 ohms?
	STEP 3C. Check for a pin-to-pin short circuit in the OEM harness.	Greater than 100k ohms?
	STEP 3D. Check for an inactive fault code.	Fault Code 131 inactive?
STEP 4.	Check ECM calibration and clear fault codes.	
	STEP 4A. Check if an ECM calibration update is available.	If a calibration update for this fault code is available, does the ECM contain that revision or higher?
	STEP 4B. Disable the fault code.	Fault code inactive?

STEP 1. Check the fault codes.

STEP 1A. Check for sensor supply fault codes.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Check for sensor supply fault codes. Use INSITE™ electronic service tool to read the fault codes.	Fault Code 2185 active? YES	Troubleshoot Fault Code 2185
	Fault Code 2185 active? NO	1B

STEP 1B. Check for an active fault code.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for an active fault code. Use INSITE™ electronic service tool to read the fault codes.	Fault Code 131 active? YES	2A
	Fault Code 131 active? NO	Use the following procedure for inactive or intermittent fault code. Refer to Procedure 019-362 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)

STEP 2. Check the accelerator pedal or lever position sensor and circuit.

STEP 2A. Inspect the accelerator pedal or lever position sensor and connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the accelerator pedal or lever position sensor from the OEM harness.

Action	Specification/Repair	Next Step
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Inspect the OEM harness and sensor connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the sensor or harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. Repair or replace the OEM wiring harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A
	Dirty or damaged pins? NO	2B

STEP 2B. Check the accelerator pedal position sensor supply voltage and return circuit.

Conditions : • Turn keyswitch OFF. • Disconnect the accelerator pedal position sensor from the OEM harness. • Turn keyswitch ON.		
Action	Specification/Repair	Next Step

Check the supply voltage and return circuit. Measure the voltage from the accelerator pedal position sensor +5 VOLT SUPPLY pin to the accelerator pedal position sensor RETURN pin at the sensor connector of the OEM harness. Use a wiring diagram for pin identification and the following procedure for general multimeter usage. Refer to Procedure 019-359 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html)	Between 4.75-VDC and 5.25-VDC? YES	2C
	Between 4.75-VDC and 5.25-VDC? NO	3A

STEP 2C. Check the circuit response.

Conditions :

- Turn keyswitch OFF.
- Disconnect the accelerator pedal position sensor from the OEM harness.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for the appropriate ECM response after 30 seconds. Use INSITE™ electronic service tool to read the fault codes.	Fault Code 132 active and Fault Code 131 inactive? YES	2D
	Fault Code 132 active and Fault Code 131 inactive? NO	3A

STEP 2D. Check the fault codes and verify accelerator pedal condition.

Conditions :

- Turn keyswitch OFF.
- Connect the accelerator pedal or lever to the OEM harness.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for the appropriate circuit response after 30 seconds. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 131 active? YES Repair : A damaged accelerator pedal or lever has been detected. Contact a Cummins® Authorized Repair Location for repair instructions. Replace the accelerator pedal or lever. See equipment manufacturer service information.	4A
	Fault Code 131 active? NO Repair : None. The removal and installation of the connector corrected the fault.	4A

STEP 3. Check the ECM and original equipment harness (OEM) harness.**STEP 3A. Inspect the ECM and OEM harness connector pins.**

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM harness from the ECM connector.

Action	Specification/Repair	Next Step
Inspect the OEM harness and ECM connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. Repair or replace the OEM wiring harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A
	Dirty or damaged pins? NO	3B

STEP 3B. Check for an open circuit in the OEM harness.**Conditions :**

- Turn keyswitch OFF.
- Disconnect the OEM harness from the ECM connector.
- Disconnect the accelerator pedal position sensor from the OEM harness.

Action	Specification/Repair	Next Step
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Check for an open circuit. Measure the resistance between the OEM harness ECM connector accelerator pedal RETURN pin and the OEM harness accelerator pedal RETURN pin. Use a wiring diagram for pin identification and the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 10 ohms? YES	3C
	Less than 10 ohms? NO Repair : An open return circuit has been detected in the OEM harness. Repair or replace the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A

STEP 3C. Check for a pin-to-pin short circuit in the OEM harness.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM harness from the ECM connector.
- Disconnect the accelerator pedal position sensor from the OEM harness.

Action	Specification/Repair	Next Step

Check for a pin-to-pin short. • Measure the resistance between the accelerator pedal SIGNAL pin in the OEM harness ECM connector and all other pins in the OEM connector. Refer to the circuit diagram or the wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES	3D
	Greater than 100k ohms? NO Repair : Repair or replace the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A

STEP 3D. Check for an inactive fault code.

Conditions : • Connect all components. • Turn keyswitch ON. • Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step

Check for the appropriate circuit response after 30 seconds. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 131 inactive? YES Repair : None. The removal and installation of the connector corrected the fault.	4A
	Fault Code 131 inactive? NO Repair : Replace the ECM. Refer to Procedure 019-031 in the Associated Procedures Table.	4A

Compare the ECM code and revision number in the ECM to the calibration revisions listed in the ECM Calibration Revision History for applicable changes related to this fault code. Use INSITE™ electronic service tool to find the present ECM code and revision number in the ECM. The ECM code and revision number are found in the Calibration Information section of System ID and Dataplate in Features and Parameters.	If a calibration update for this fault code is available, does the ECM contain that revision or higher? YES	4B
	If a calibration update for this fault code is available, does the ECM contain that revision or higher? NO Repair : If necessary, calibrate the ECM. Refer to Procedure 019-032 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)	4B

STEP 4B. Disable the fault code.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Disable and clear the fault code. Operate the engine within the "Conditions for Clearing the Fault Code" found in the Overview section of this troubleshooting procedure.	Fault code inactive? YES	Repair complete
	Fault code inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

Associated Procedures

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/336/336-019-031.html)	ISL8.9 CM2880 L112	4358493
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/164/164-019-031.html)	ISL9.5 CM2150 SN	4310608
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/209/209-019-031-tr.html)	ISB3.9 CM2220 B107	4310792
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/218/218-019-031.html)	QSB4.5 CM2150 B108	4326163
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/216/216-019-031.html)	QSB6.7 CM2150 B109	4326168
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/331/331-019-031.html)	QSB7 CM2880 B117	4358390
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/177/177-019-031-tr.html)	ISG11 CM2880 G106	4332695

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/177/177-019-031-tr.html)	ISG12 CM2880 G107	4332690
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/325/325-019-031-tr.html)	ISG11 CM2880 G108	4332901
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/325/325-019-031-tr.html)	ISG12 CM2880 G109	4332906
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/206/206-019-031-tr.html)	QSNT14 CM876 N102	4325993
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/284/284-019-031-tr.html)	QSF2.8 CM2880 F104	4332741
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/284/284-019-031-tr.html)	QSF2.8 CM2880 F108	4332746
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/389/389-019-031-tr.html)	QSF3.8 CM2880 F112	4383825
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/383/383-019-031-tr.html)	ISB5.9 CM2880 B127	4383645

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/379/379-019-031.html)	ISB/ISD6.7 CM2880 B126	4383693
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/403/403-019-031.html)	QSL9.3 CM2880 L113	4383811
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/387/387-019-031.html)	ISF3.8 CM2220 F116	4383664
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/436/436-019-031.html)	QSC8.3 CM2880 C102	4388785
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/451/451-019-031.html)	QSB5.9 CM2880 B139	4388870
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/449/449-019-031.html)	QSB3.9 CM2880 B138	5411050
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/506/506-019-031.html)	ISD6.7 CM2880 D101	5411372
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/497/497-019-031.html)	ISF2.8 CM2220 F129	5411325

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/501/501-019-031.html)	ISF4.5 CM2220 F123	5411320
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/556/556-019-031.html)	ISF3.8 CM2220 F134B	5504165
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/577/577-019-031.html)	Z14 CM2670 Z103B	5504577
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/546/546-019-031.html)	D6.7 CM2670 D102B	5504515
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/629/629-019-031.html)	B6.2 CM2670 B156B	5579510
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/624/624-019-031.html)	X12 CM2670 X121B	5504455
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/622/622-019-031.html)	L9 CM2670 L128B	5504589